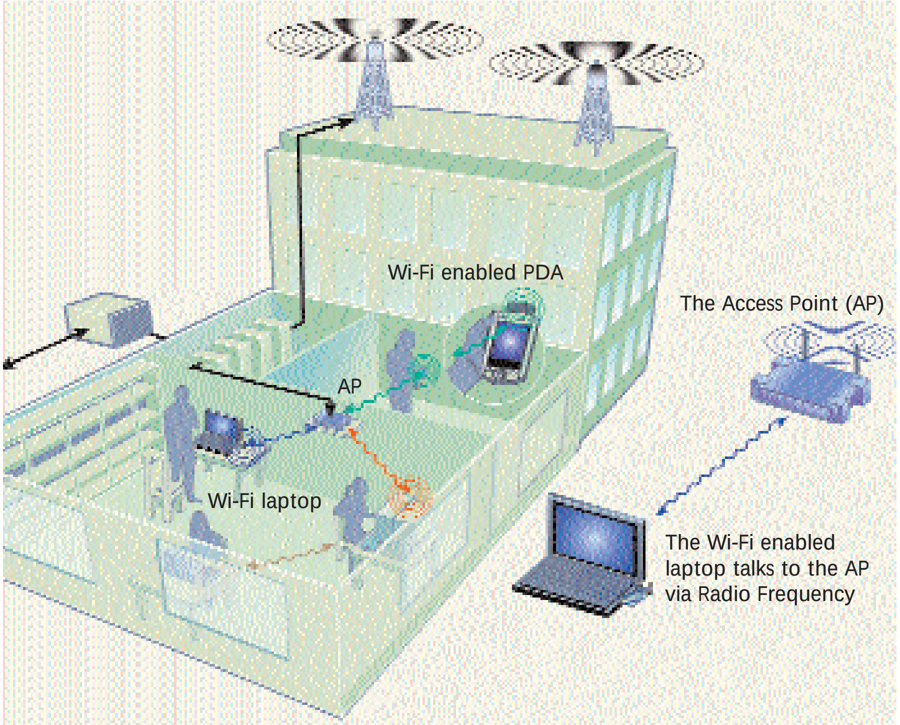
instantly provided the necessary fixes. Giri says, "A service outage could mean a faulty AP, or problems in the wired infrastructure, or lack of connectivity at the ISP's end."

Convergent's technical team checks the LRM APs on a daily basis for all quality-of-service parameters, and confirms that the bandwidth available is upto the mark. Pathways didn't have any major service outages either. In Jain's words, their network is up, "working fine, without any congestion".

Maintenance, too, is not a problem per se. Jain says that their in-house tech staff, though "not experts", can easily field small problems. Shankar points out, "Wi-Fi is new technology around here, and everyone is eager to push it. Still, for a really crowded network, the 11 Mbps throughput may not be quite adequate, and never as good as a wired LAN".

Wired LANs still give you higher throughput than a Wi-Fi LAN. The maximum speeds on a wired network are 100 Mbps, and 11 Mbps for a 802.11b Wi-Fi. In practice, wired LANs give about 50-60 Mbps and about 5-6 Mbps is considered fine for Wi-Fi networks. While normal Internet access isn't really affected by this, heavy LAN usage, such as accessing large files on the LAN, will be noticeably slower on a Wi-Fi network.

D-Link's General Manager for Development and Product Support, Shridhar Kadam lists the reliability and reach of WLANs as important parameters that make them popular, and goes on to say that with both wired and wireless LAN generally co-existing within the same environment, its important to ensure that the two are compatible and manageable as one network.



Cost per PC for setting up a wired and a wireless office LAN

WIRED		WIRELESS	
1 10/100 Mbps LAN card	Rs 500	1 PCMCIA Wi-Fi card	Rs 2,800
CAT5 cable 20m length	Rs 200	1/20th access point	Rs 250
1/24th Patch Panel port, information outlets, & other hardware	Rs 200		
GRAND TOTAL:	**Rs 900**		**Rs 3,050**

The cost of switches will be common to both; wireless LAN will be a 11/22 Mbps (using standard 802.11b or proprietary 802.11b+ technology). The wired LAN will have a 100 Mbps throughput.

All costs are approximate

Jargon Buster

802.11 standards: The 'b' standard came first, operating on the 2.4 GHz spectrum, offering 11 Mbps throughput. Next up was 802.11a that operated at 5 GHz, with a 54 Mbps speed. This was quickly superseded by the 802.11g that operates on the 2.4 GHz band and gives 54 Mbps.
Access points: Wi-Fi enabled desktops and laptops communicate via radio frequency with the Access Point (AP). The AP then connects to the backend servers that provide the necessary infrastructure to connect to the Internet.
WPA: Wi-Fi Protected Access, a security protocol suite that has replaced WEP, using a complex, and until now, secure system of authentication.
WEP: Wireless Equivalent Privacy, an older security protocol that was used widely in older APs, to provide security. WEP is now considered relatively insecure.

How far would they go

So impressed was Jain with the Wi-Fi network at the Pathways campus, that he implemented one at home. Exel's Subramaniam says that eventually, he wants to get entire new offices go to Wi-Fi. "It's not tough to set it up", is the common refrain. As of now, Wi-Fi adoption in India is driven by corporate adoption and the recent fall in laptop prices. Typically, laptop users are those in higher management and well, you do need to be that highly placed to shell out Rs 70,000 to buy a decent laptop!

Undoubtedly, Wi-Fi networks are going to spread within a few years, and will probably be commonplace within the next five years. As hardware costs keep coming down, expect more activity on the wireless front. While Wi-Fi will never replace conventional wired access, it will provide access at places where we couldn't get connected earlier.

Don't bet too much on getting cheap public hotspot access, though. Sify charges about Rs 515 for 8 hours of Wi-Fi connectivity in Bangalore.

Appasamy says very matter-of-factly, that unlike in the US, "The average Joe who walks into a coffee bar is not going to carry a laptop", indicating that wire-free Internet access is still going to be firmly targeted at the individuals who are willing to pay, for it. At LRM, they discreetly tell us that guests are charged Rs 500 for 24 hours of access, or Rs 250 for every hour of connectivity.

Nevertheless, the luxury of sitting by the sunny poolside, sipping chilled water and surfing the Internet makes it look reasonable. Almost!

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